

YASHAS CHANDRA BATHINI AI / ML Engineer | LLMs | AWS | RAG | FastAPI | Full-Stack Development

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 yashas144.github.io/AI-Engineer-Portfolio  F-1 OPT | Authorized to Work in the U.S.

PROFILE

AI/ML Engineer with 4 years of experience designing and deploying Agentic AI systems, LLM-powered applications, MLOps pipelines, distributed microservices, and cloud-native platforms. Experienced in autonomous AI agents, Retrieval-Augmented Generation (RAG), real-time analytics, and scalable API architectures. Proven expertise building production-grade AI solutions using Python, Java, FastAPI, React, AWS and Azure. Strong background in event-driven systems, cloud deployments, MLOps, and high-performance AI apps serving mission-critical business operations.

SKILLS

Generative AI & LLMs: Transformers, Hugging Face, RAG, Agentic AI, LangChain, MCP, Crew AI, Semantic Kernel, NLP, FAISS, ChromaDB.

Full-Stack Web Development: React, Spring Boot, Microservices, FastAPI, Flask, REST APIs, Webhooks, LLM Integration.

Programming: Python, Java, OOP, Design Patterns, PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Cursor, Copilot.

MLOps: MLflow, Experiment Tracking, Model Registry, Model Monitoring, Model Versioning, AI Evaluation

Cloud & DevOps: AWS (S3, Lambda, SageMaker, Bedrock), Azure, Docker, Git, GitHub Actions, CI/CD pipelines.

Data Engineering: ETL/ELT, Data Pipelines, Event-Driven Architecture, Apache Spark, PySpark, Databricks, Tableau, MySQL, PostgreSQL.

WORK EXPERIENCE

AI/ML Engineer 09/2020 – 08/2024
Sysarket Hyderabad, India

- AfterHours AI - AI Voice Booking Orchestration System**
- Led the design and delivery of a conversational AI booking system, integrating a **Trillet AI** voice chatbot agent with a FastAPI **webhook handler** to automate appointment capture for a **podiatry practice** with **zero manual data entry**.
 - Engineered a **parallel async pipeline** and **WebSocket broadcast layer**, streaming live booking updates to a **React dashboard** in **under one second** for **10,000+ daily events**.
 - Designed an **idempotency layer** (event-ID deduplication) across Google Calendar and Sheets APIs to prevent duplicate records on webhook retries.
 - Validated **webhook payloads** with **Pydantic** at the API boundary to catch malformed data before processing.

- Auravance – Industrial Predictive Intelligence Platform**
- Built a full-stack AI platform (React, FastAPI) for **real-time equipment monitoring** and **failure prediction** using XGBoost and LSTM, improving detection **accuracy by 30%**.
 - Developed a RAG-based **root cause analysis system** over equipment manuals, generating **explainable diagnostics**.
 - Designed and optimized **ETL pipelines** processing **1M+ sensor records/day** into analytics-ready datasets for predictive maintenance, anomaly detection, and RUL models, enabling **sub-minute insights** and reducing **equipment downtime by 20%**.
 - Engineered scalable event-driven **IoT ingestion pipelines** using **MQTT** and **Python** for high-frequency sensor telemetry with **low latency** and **high reliability**.

- AI-Powered Music Intelligence Platform**
- Designed and deployed an AI-powered music recommendation platform leveraging **Hybrid-RAG (BM25 + vector search)** across **1M+ songs**, enabling context-aware discovery based on **mood, genre, and user intent**.
 - Architected a scalable RAG pipeline with **embedding-based retrieval** and **reranking**, significantly improving recommendation relevance and reducing noise in outputs.
 - Engineered real-time features, including **real-time mood detection** and **seamless playback**.
 - Deployed a cloud-native full-stack system (React + FastAPI) on **AWS** and **Hugging Face** with **CI/CD pipelines**, ensuring reliable and scalable delivery.

- FIFA Analytics & Prediction System**
- Developed a robust sports analytics platform featuring ML-based **match predictions, APIs, and dashboards**.
 - Improved **model accuracy by 18%** using feature engineering, hyperparameter tuning, and **MLflow tracking**.
 - Implemented model registry, versioning, and monitoring (performance, confidence, and feature drift).
 - Deployed a containerized **full-stack application** using React (frontend) and Java (backend) with **integrated ML services**, leveraging **Docker** and **Azure** with **Azure SQL DB** for scalable, production-ready model serving.

- Deep Learning-Based Image Synthesis with GAN Architecture | Research Publication**
- Developed GAN-based image synthesis models using **PyTorch** and **TensorFlow**, improving output quality.
 - Contributed to **generative AI research** by designing GAN architectures for real-world applications.

EDUCATION

Master of Science in Data Science Denton, Texas
University of North Texas, CGPA: 4.0 / 4.0

Bachelor of Technology in Information Technology Hyderabad, India
Jawaharlal Nehru Technological University Hyderabad, CGPA: 3.7 / 4

CERTIFICATIONS

Generative AI Practitioner - AWS | Generative AI Fundamentals - Databricks | Machine Learning with Big Data - UC San Diego | Introduction to Data Science - Cisco | Career Essentials in Generative AI - Microsoft | Python Essentials - Cisco